MILITARY SPECIFICATION
FLIGHT COORDINATION CENTRAL AN/TSC-61( )

1. SCOPE

1.1 This specification covers one type of Army Air Traffic Regulator System which provides extension capabilities for Flight Operations Centrals and is designated Flight Coordination Central AN/TSC-61( ). (See 6.1 and 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-13231	Marking of Electronic Items
MIL-P-11268	Parts, Materials and Processes Used In Electronic Communication Equipment
MIL-F-14072	Finish for Ground Signal Equipment
MIL-S-19500	Semi-conductor Devices, General Specification For
MIL-I-46058	Insulation Compound, Electrical (For Coating Printed Circuits Assemblies)
MIL-P-55110	Printed Wiring Boards
MIL-S-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging and Packing Of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

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MILITARY (Cont'd)

MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-633	Electric Power Engine Generator Set Family Characteristic Data Sheets
MIL-STD-810	Environmental Test Methods
MIL-STD-1332	Definitions of Tectical, Prime, Precise, and Utility Terminologies for Classification of DOD Mobile Electric Power Engine Generator Set Family

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 System description. The Flight Coordination Central (FCC) is primarily a communication facility utilized in the Army Manual Air Traffic Control System. It provides the ground-air-ground and ground-to-ground communications equipment and other accessory items required for coordination of the Air Traffic Control functions. The system usually functions as an extention of Flight Operation Centrals (FOC) which are the main traffic control facilities of the enroute system.

3.2 Procurement model. A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model and the equipment shall incorporate all features of the model.

NOTE: In case of conflict between specified performance characteristics for the equipment and performance of the model, the specified performance characteristics govern.

3.3 First Article Samples. - The contractor shall furnish First Article samples of equipment on order as required by the invitation for bids and contract. (SEE 4.3 & 6.2)

3.4 Cleaning. -

3.4.1 Parts. - After fabrication, parts shall be cleaned as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive meterials shall be removed completely before the parts are mounted on chassis, panels, etc.

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3.4.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.6 Finish, Protective. - The equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc. and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be the same color as the procurement model applied in accordance with and conforming to MIL-F-14072.

3.7 Securing of parts. -

3.7.1 General. - Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.7.2 Securing of parts by threaded fasteners. - Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.8 Marking. -

3.8.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be in accordance with the equipment drawings.

3.8.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.8.3 Serial Numbers. - The unit(s) which are contractor-furnished and assigned an approved nomenclature and the AN/TSC-61( ) shall be serial numbered.

3.9 Soldering. -

3.9.1 Solder. - Solder shall conform to Requirement 5, of MIL-STD-454.

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3.9.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Wiring and cabling. - Wiring and cabling shall be neat and sturdy.

3.10.1 Cabling. - Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance.

3.10.2 Slack. - Wires and cables shall be as short as practicable except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the part to which the wire or cable is connected, except where uninsulated wires are used.

3.10.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

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3.10.4 Support. - Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner.

3.10.5 Clearance. - Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.10.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.10.7 Connections, general. - Before being soldered to terminal lugs or fixed terminals, wire shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to complete electrical connections.

3.11 Printed wiring assemblies (SEE 4.4). - Printed wiring assemblies shall be processed for compliance with the procurement model and MIL-P-55110 in order not to affect life and serviceability of the assemblies.

3.12 Conformal coatings (SEE 4.4). - Except as otherwise indicated on the procurement model, printed wiring assemblies shall be conformally coated with a coating material which conforms to MIL-I-46058, Type ER (Epoxy) or Type PUR (Polyurethane). The coating shall be applied to both sides of the cleaned printed wiring assembly including the component leads. These assemblies shall be cleaned of flux and other contaminants prior to application of coating. Cleaning compounds shall have no deleterious effects on any part of the printed wiring assembly. The conformal coating shall be compatible with all parts of the printed wiring assembly and the thickness shall be 0.0035 inch nominal tolerance. The coated assemblies shall show no visible bubbling, blistering, wrinkling, cracking, pinholes, or peeling of the coating material or corrosion of printed conductors. Printed wiring assemblies shall be adequately masked or otherwise protected in such a manner that application of conformal coating does not degrade the electrical performance of the assembly. Assemblies having adjustable components shall not have the adjustable portion covered with the coating. Electrical and mechanical mating surfaces such as connector contact points, test points,

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screw threads, bearing surfaces, etc., shall not be coated. When masking those areas which are not to be coated, and tape is used, the tape shall be an electrical grade. When the printed wiring assembly utilizes glass encased components, such components shall be protected against breakage by the conformal coating by covering the component with thin pliant buffer material, prior to coating, such as polyvinylidene fluoride or silicone rubber. This buffer material shall be compatible with the conformal coating material and all parts of the printed wiring assembly. In addition, the buffer material shall be fungus and flame resistant; and shall be clear transparent so markings on the component shall be visible. Conformal coatings shall be clear transparent so markings on the components shall be visible.

3.13 Semiconductors (transistors & diodes) (SEE 4.4). - These electron devices (semiconductors and diodes) shall conform to Specification MIL-S-19500, and procurement model.

3.14 Power. -

3.14.1 Prime power. - The equipment as a system shall be capable of operating from GFE military standard generator sets having characteristics listed in MIL-STD-1332 for Type I, Class 2, Mode III generator sets. Power unit assemblages utilized shall contain appropriate generator sets listed in MIL-STD-633. The equipment shall also be capable of operating from commercial 3 phase power 120/208 volts $\pm$ 10%, 50/60 Hz $\pm$ 5%.

3.14.2 AC power transfer. - A power transfer switch shall be provided to facilitate connection of two power sources to the shelter and allow connection of both to the main power bus but not simultaneously. This transfer switch shall be located in the shelter. A system of lights shall be utilized to indicate generator set furnishing power.

3.14.3 Load unbalance. - The load unbalance for any one phase of the three phase power system shall not exceed by 20 percent the average current load of either of the other two phases.

3.14.4 Power protection. - Protection devices shall protect all equipment against transient voltages and variations in the prime power source which include the following:

a. Over-voltage line surges and transient voltage spikes.

b. Transient voltages of 600v peak for a period of one millisecond on each phase.

c. Under-voltage below the minimum specified.

d. Loss of input phase.

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3.14.5 Power Phase sequence device. - This device shall indicate by visual means when phase sequence of the prime power phase voltage is incorrect, and, shall also not permit closure of the main circuit interrupter.

3.14.6 DC power. - The power supplies shall be in accordance with the procurement model as modified by the contract.

3.14.7 Automatic dc changeover. - During normal operation and with all dc powered equipment operating, the load is divided between two power supplies. In the event of failure of either supply, the load will automatically be switched to the remaining one.

3.15 Provision for storage of operation components during transit. - Provisions shall be made within the shelter for storing the antenna system, and its associated hardware and cables during transit. The provisions for securing the stored items during transit shall be designed so as to offer no obstacles to personnel in performing their duties.

3.16 Controller communications facilities. - The two controller positions and the supervisor's position have identical and interchangeable sets of communications control equipment. Each position has a Radio Key Panel, a Telephone Key Panel and Audio Control Box.

3.16.1 Radio key panel. - The Radio Key Panels have ten 3 position (locking each position) selector key switches, ten indicator lamps. The switches each have an "off" position-down, a "monitor" position-center and a "push to Xmit" position-up. The indicator lamps flash when the message is received and the associated switch is in the "off" or "monitor" position. The indicator lamps shall be steady on when the associated selector switch is in the "push to Xmit" position.

3.16.2 Telephone key panel. - The telephone key panel shall have fifteen selector key switches and fifteen indicator lamps. Each selector switch shall have three positions: "ring"-down position (non-locking), "off"-center position (locking) and "talk"-up position (locking). The indicator lamps shall flash on an incoming call when the associated switch is in the "off" position. The lamps shall be steady on when the associated switch is in the "talk" position.

3.16.3 Audio control box. - The audio control box contains the audio amplifier circuits, hybrid coil, volume controls, and provide the microphone headset output inputs. It also includes the switching circuit required to allow the routine of telephone and radio audio.

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3.16.4 Switch in A+B position. Radio audio is routed to the headset microphone, connected to the "A" operator cable (CX-2556/U). The telephone audio selected is routed to the headset microphone connected to the "B" operator cable (CX-2555/U).

3.16.5 Switch in A position. Radio audio selected at the associated radio key panel is routed to the "A" operator cable (CX-2556/U) as above. However, selection of a telephone line also routes the telephone audio to the "A" cable and deactivates the radio transmitter circuit. This allows the "A" operator to use the telephone while monitoring the radio receiver and prevents keying and modulating a transmitter at the same time. Under this condition cable "B" is inoperative.

3.17 Signal control box. The signal control box shall provide the interface circuits between the radios and controller consoles and house the telephone circuits.

3.17.1 Radio control printed circuit card. This card shall provide a termination and level adjustment for the associated receiver, a circuit to sense a receiver output and light the associated indicator lamps at all key panels and a microphone circuit input for isolation and level adjustment.

3.17.2 Telephone control printed circuit. The telephone control printed card shall provide a circuit to sense an incoming ringing signal and cause the associated lamp to flash at each controller position. The circuit shall also switch telephone audio to the appropriate line when the telephone key switch is placed in the "talk" position and shall switch the ringing signal to the appropriate line when the key switch is depressed to the "ring" position.

3.17.3 Buzzer and flasher card. The buzzer and flasher card shall control the indicator lamp and the telephone night alarm buzzer circuit.

3.18 Voice communications system performance. The voice communications system shall meet the following requirements:

3.18.1 UHF AM channels. The UHF AM radio channels shall meet the following system requirements:

3.18.1.1 Audio output (receiver). The audio output measured at the headset output of the audio control box shall be 230 mW minimum, with an RF input at the antenna entrance panel of 1000 microvolts, modulated 30% and at a modulating frequency of 1000 Hz.

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3.18.1.2 Sensitivity (receiver circuit). An RF input to the antenna entrance panel of 4 microvolts modulated 30% with a 1000 Hz signal shall produce a signal to noise plus noise ratio of at least 10 dB when measured at the audio control box headset output.

3.18.1.3 Distortion (receiver). The distortion shall be 12% or less with an RF input to the antenna entrance panel of 1000 microvolts modulated 30% at 1000 Hz and with the output level at the audio control box set to 100 milliwatts.

3.18.1.4 VSWR (transmitter). The VSWR shall be 3.0 to 1 maximum over the entire frequency range when the connector on the antenna entrance panel is terminated with the system cables and Antenna AT-197/GR.

3.18.1.5 Power output (transmitter). The output power at the antenna entrance panel shall be 14 watts minimum.

3.18.1.6 Percent modulation (transmitter). With 0.6 millivolts at 1 kHz to the microphone input of the audio control box the modulation shall be 75% minimum without clipping at the antenna entrance panel.

3.18.1.7 Sidetone (transmitter). With an input of 0.6 mV at 1 kHz to the audio control box, there shall be a sidetone present at the speaker panel.

3.18.1.8 Distortion (transmitter). With 0.4 millivolts at 1 kHz applied to the microphone input of the audio control box, the distortion measured at the antenna entrance panel shall be 10% maximum.

3.18.2 VHF AM channels. The VHF circuits of the central shall meet the following requirements:

3.18.2.1 Audio output (receiver). The audio output measured at the headset output of the audio control box shall be 230 mW minimum, with an RF input at the antenna entrance panel of 1000 microvolts, modulated 30% and at a modulating frequency of 1000 Hz.

3.18.2.2 Sensitivity (receiver). An RF input to the antenna entrance panel of 3 microvolts modulated 30% by a 1000 Hz signal shall produce a signal + noise to noise ratio of at least 6 dB measured at the audio control box headset output.

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3.18.2.3 Distortion (receiver). The distortion shall be 12% or less with an RF input to the antenna entrance panel of 1000 microvolts modulated 30% at 1000 Hz and with the output level at the audio control box set to 100 milliwatts.

3.18.2.4 Power output (transmitter). The output power at the antenna entrance panel shall be 20 watts minimum.

3.18.2.5 VSWR (transmitter). The VSWR shall be 3.0 to 1 maximum over the entire frequency range when the connector on the antenna entrance panel is terminated with the system cables and Antenna AS-112/TRC-7.

3.18.2.6 Percent modulation (transmitter). With 0.6 millivolts at 1000 Hz applied at the microphone input of the audio control box the modulation shall be 75% minimum without clipping.

3.18.2.7 Sidetone (transmitter). With 0.6 millivolts at 1000 Hz applied at the microphone input to the audio control box, there should be a 50 mW to 100 mW sidetone level (volume control maximum) present at the headset output.

3.18.2.8 Distortion (transmitter). With 0.6 millivolts at 1000 Hz applied to the microphone input of the control panel, the distortion measured at the antenna entrance panel shall be 10% maximum.

3.18.3 VHF-FM channels. The FM circuits of the central shall meet the following requirements:

3.18.3.1 Audio output (receiver). The audio level measured at headset output of the audio central box shall be 230 mW minimum, with an RF input at the antenna entrance panel of 1000 microvolts, and $\pm$ 8 kHz deviation at 1000 Hz.

3.18.3.2 Sensitivity (receiver). An RF input signal of 1.5 microvolt deviated $\pm$ 8 kHz applied at the antenna entry shall produce a signal + noise to noise ratio of at least 10 dB when measured at the headset output of the radio control box.

3.18.3.3 Distortion (receiver). The distortion measured at the headset output of each audio control box shall be 10% or less when an RF input of 1000 microvolts deviated $\pm$ 8 kHz by a 1000 Hz signal is applied to the antenna entry panel and the audio control box "radio" volume control is set for 100 mW.

3.18.3.4 Power output (transmitter). The output power at the antenna entrance panel shall be 20 watts minimum.

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3.18.3.5 VSWR (transmitter). The VSWR shall be 3.0 to 1 maximum over the entire frequency range when the connector on the antenna entrance panel is terminated with the system cables and Antenna AS-1729/VRC.

3.18.3.6 Deviation (transmitter). With 0.6 millivolts at 1000 Hz applied at the microphone input of the audio control box, the deviation shall be from ± 6 kHz to ± 10 kHz.

3.18.3.7 Sidetone (transmitter). With 0.6 millivolt, 1000 Hz signal applied to microphone input of the audio control box there shall be a sidetone level of between 25 mW to 140 mW present at the headset output (volume centrol maximum).

3.18.3.8 Distortion (transmitter). The distortion of detected audio measured at the antenna entry panel shall be less than 10% when a 1000 Hz signal of 0.6 millivolts is applied to the microphone input of the audio control box.

3.18.4 HF-SSB channel. The HF-SSB circuits shall meet the following requirements:

3.18.4.1 Audio output (receiver). The audio output level measured at the audio control box headset output shall be 230 mW minimum with an input of 1000 microvolts modulated 30% with a 1000 Hz signal applied to the R-T unit Antenna input connector.

3.18.4.2 Sensitivity (receiver). An RF input of 3 microvolts modulated 30% with a 1000 Hz signal applied to the antenna connector on the R-T unit shall produce a signal + noise to noise ratio of at least 10 dB measured at the audio control box headset output.

3.18.4.3 Power output (transmitter). With the transmitter function switch turned to AM the power output measured at the antenna entrance panel shall be 100 watts minimum.

3.18.4.4 Percent modulation. With a 0.6 millivolt 1000 Hz signal applied to the microphone input of the audio control box the modulation shall be 75% minimum.

3.18.4.5 Sidetone (transmitter). With a 0.6 millivolt 1000 Hz signal applied to the microphone input of the audio control box the sidetone measured at the headset output shall be between 230 mW maximum and 100 mW minimum.

3.18.5 Radio channel crosstalk. Crosstalk isolation between radio channels shall be a 50 dB minimum.

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3.18.6 Volume control attenuation. Radio channel volume control attenuation shall be 40 dB minimum when rotated from full-on to full-off.

3.18.7 Foot switch. Each audio control box will have connection for a foot operated switch to key selected radio transmitter.

3.18.8 Voice security.

3.18.8.1 General. Provisions shall be included to secure one FM radio channel using available ground type remote controlled tactical voice security equipment. All security equipment shall be physically and electrically isolated from the remaining communication equipment by use of shielded twisted pair wiring.

3.18.8.2 Electrical. A 100 watt, 28 vdc source from one of the power supplies shall be available for operation of the security equipment.

3.18.8.3 Physical. In a suitable location as in the model, a space at least 18 inches high, 18 inches wide, and 18 inches deep shall be provided for mounting security equipment weighing 50 pounds. A clear space of at least 36 inches shall be provided in front of the security equipment for frequent mounting and removal.

3.18.8.4 Amplification. The amplifier portion of a C-1611 communication control set shall be used in conjunction with the secure channel to provide microphone and earphone amplification. It shall also provide volume control for the earphone level.

3.18.8.5 Cross talk. Analog traffic being received or transmitted on the secured set shall not appear as cross talk on any other circuit or set at a level greater than 100 dB down from the intended level, with all levels set so as to simulate the system in normal operation and with all signals properly terminated as in normal operation.

3.18.8.6 FM receiver/transmitter Nr. 3. It shall be possible to disconnect FM receiver/transmitter Nr. 3 from the secured circuit and reconnect it to the main radio control system where it can be utilized as one of the three available FM receiver/transmitters.

3.18.9 Telephone system requirements. Each telephone line shall meet the following requirements:

3.18.9.1 Received audio level. With an input of zero (0) dBm 1000 Hz at the telephone entry panel binding posts the output at the audio control box headset output shall be 25 mW minimum (volume control maximum).

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3.18.9.2 Transmitted audio level. With a signal of 0.6 millivolt 1000 Hz applied to the microphone input of the audio control box the output at the telephone entry panel binding posts shall be between -6 to +3 dBm.

3.18.9.3 Crosstalk (telephone). Crosstalk isolation between telephone lines shall be at least 40 dB.

3.18.9.4 Night alarm buzzer. A buzzer shall be located in the operations room which will function when a telephone ringing signal is received. The buzzer will cease when the call is answered. A switch will be provided in the signal control box to disconnect the buzzer.

3.18.10 Teletype system. The teletype shall be capable of operation by either radio or land lines.

3.18.10.1 Land Line operation. The land line teletype shall operate as a teletype only, local battery system, connected for two wire operation. Integration of the various elements shall not degrade the operation of any one of them.

3.18.10.2 Radio teletype operation. For radio-teletype operation, the "Land line-radio" switch is placed in the "radio" position. In this position, the teletype send and receive circuits are connected in series with the 700B-2 teletypewriter converter oscillator located in the 718F-1 transceiver case.

3.18.10.2.1 Radio teletype performance. The radio teletype set shall meet the following requirements:

3.18.10.2.2 Transmitter output. The transmitter power output shall be 100 watts average minimum.

3.18.10.2.3 Receiver sensitivity shall be at least 10 dB signal + noise to noise for a 3 microvolt RF input signal.

3.18.10.2.4 The radio teletype shall be set for 60 words-per-minute (wpm) 368.1 operations-per-minutes (opm). Accessory gears are included for operation at 100 wpm (600 opm).

3.19 Service conditions. The Flight Coordination Central shall be designed and constructed to comply with the following environmental requirements:

a. Altitude. The contractor-fabricated internal equipment and assembly shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level and shall withstand air transportation at altitudes up to 50,000 feet above sea level.

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b. High temperature.

(1) The assembly shall be operable without degradation in specified performance at ambient air temperatures as high as 125°F plus the effects of solar radiation at a rate of 360 BTU/ft 2/hr. The assembly shall be permitted a 30 minute cool-down period before full operation is required. During the 30 minute period, the assembly ventilating system(s) (not air conditioning) and the electronic equipment may be turned on as desired to benefit overall operational performance. At the end of the 30 minute period, the assembly shall provide full specified performance.

(2) Storage and transportation. The contractor-furnished internal equipment and the assembly shall withstand exposure to ambient air temperatures as high as +160°F.

c. Low temperature.

(1) Operating. The assembly shall be operable without degradation in specified performance at ambient air temperatures as low as -50°F. The assembly shall be permitted a 30 minute warm-up period before full operation is required. During the 30 minute period, the assembly heating system(s) and the electronic equipment may be turned on as desired to benefit overall operational performance. At the end of the 30 minute period, the assembly shall provide full specified performance.

(2) Storage and transportation. The contractor-fabricated internal equipment and the assembly shall withstand exposure to ambient air temperature as low as -70°F.

d. Humidity. The contractor-fabricated equipment and the assembly shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to relative humidities as high as 100 percent at all ambient air temperatures to 80°F, high absolute humidity corresponding to a dew point of 86°F, and low relative humidity of 5 percent at 125°F.

e. Rain. The assembly shall be operable without degradation in specified performance, and shall sustain no physical damage, during precipitation at variable rate to a total of 9.50 inches over a 12 hour period with intermittent wind speed to 35 knots.

f. Wind. The assembly shall be capable of withstanding winds of 52 mph for a 5-minute period with gusts to 75 mph. Hold down facilities shall be provided if required to withstand winds of 65 mph for a period of 5-minutes with gusts to 100 mph. A shelter assembly shall meet the requirement in vehicle mounted condition.

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g. Sand and dust.

(1) The assembly shall withstand, in both operating and non-operating conditions, exposure to fine sand and dust particles with wind speeds of 35 knots.

(2) The contractor-furnished internal equipment shall withstand, in both operating and non-operating condition, the effects of blowing fine sand and dust particles with wind speeds of 35 knots surrounding the mobile enclosure containing the equipment, and shall be resistant to dust that may accumulate within the assembly enclosure as a result of operator activity.

h. Fungus. The contractor-furnished internal equipment and the assembly shall withstand exposure to fungus growth as encountered in tropical climates.

i. Water fording. Modifications to the shelter as may be required in the construction of an assemblage shall in no way reduce the inherent water fording capability of the enclosure. This capability shall be as specified in the individual shelter or van specification.

j. Shock and vibration. The assembly shall withstand vibration and shock induced during vehicular transport over all types of roads and cross country terrain. Individual contractor-fabricated electronic units shall be free of damaging resonances at frequencies below 55 cps. The assembly shall also withstand shock and vibration associated with shipment by rail, water, or air.

k. Bench handling. The contractor-fabricated electronic items shall withstand shocks encountered in servicing.

3.20 Electromagnetic compatibility (See 4.8). The complete Central TSC-61( ) shall meet the requirements of MIL-STD-461 as specified.

3.20.1 System compatibility (See 4.8.1). The total system with all equipments operating and performing their intended functions shall comply with the following MIL-STD-461 requirements:

CE03	CS06
CS02	RE03 (relaxed by 20 dB)

3.20.2 Contractor Fabricated Equipment (See 4.8.2). Contractor fabricated equipment and vendor supplied items shall comply with the applicable class requirements of MIL-STD-461.

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3.20.3 Intra-System compatibility (See 4.8.3). The complete Central TSC-61( ) with all equipments operating as required shall comply with the Intra-System EMI/EMC requirements as specified in the contract.

3.21 Parts, materials, and processes (See 4.4). In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.22 Interchangeability (See 4.9). Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the units.

3.23 Technical literature and spare parts. Technical literature and spare parts shall be furnished as specified in the contract. Spare parts furnished shall be identical to corresponding parts in the equipment furnished on order. (See 6.2)

3.24 Workmanship (See 4.10). The equipment shall be manufactured and assembled in accordance with the Procurement Model (See 3.2), and drawings procured on contract, and the following requirements and paragraphs herein:

MIL-STD-454:

Soldering	Requirement 5
Workmanship	Requirement 9

Paragraphs herein:

Cleaning	3.4
Control	3.5
Securing of parts	3.7
Soldering	3.9
Wiring & Cabling	3.10

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Contractor product assurance. The contractor shall provide and maintain a means of determining product conformance in accordance with the requirements specified herein or elsewhere in the contract. The Government, at its option, may perform any evaluation deemed necessary to assure the adequacy of the means employed and the effectiveness of the contractor's product assurance methods and systems.

4.1.2 Government verification. All quality assurance operations required of the contractor shall be subject to Government verification at scheduled or unscheduled intervals. Verification will consist of the following:

a. Surveillance of the contractor's operations to determine that practices, methods, and procedures of the contractor's product assurance system are being applied and are, in fact, accomplishing the objectives of total compliance of the product with the requirements of this specification.

b. Government product inspections to determine compliance of the product with the specification requirements.

4.1.3 Accommodation and assistance. The Government Quality Assurance Representative shall have the right of access to any area of the contractor's or his subcontractor's premises where any part of the work is being performed. The Government Quality Assurance Representative shall be afforded unrestricted opportunity to verify conformance of the product with specification requirements. The contractor shall make his inspection equipment available for use by the Government Quality Assurance Representative for verification purposes. The contractor's personnel shall be made available for operation of such inspection equipments as required.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First Article Inspection. (Does not include preparation for delivery.) (SEE 4.3)

b. Inspections covered by subsidiary documents. (SEE 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (SEE 4.5)

(2) Quality conformance inspection of preparation for delivery. (SEE 4.11)

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4.3 First Article Inspection. This inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for Group A, Group B and Group C (SEE TABLES I, II and III respectively). Environmental and stress inspections shall be performed in the order shown in TABLE IV according to the number of samples specified.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation of delivery:

ITEM	WHERE REQUIRED
Finish	3.6
Marking	3.8
Printed Wiring Assemblies	3.11
Conformal Coating	3.12
Parts, Materials and Processes	3.21
Semi-conductors (Transistor & Diode)	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification shall be made available by the contractor. A defect is any nonconformance of unit of product with the specified requirement. Any unit of product found defective during inspection is rejected. Rejected units may be repaired or corrected and then resubmitted for inspection.

4.5.1 Group A inspection. Each unit of equipment shall be inspected for conformance to all inspection and test requirements of TABLE I of this specification. In addition, equipment conforming to the above requirements shall be verified by a quality audit. The following procedure shall be used for the quality audit:

At the start of this plan 100% of the units are inspected until five consecutive units are found free of defects. At that time inspection is reduced to one out of five of the units presented. Whenever a defective unit is found, immediately revert to 100% inspection, for all attributes which is then continued until five consecutive units are found free of defects. At that time inspection is again reduced to one out of five of the

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units presented. Samples shall be selected on a random basis. The contractor shall simultaneously notify the Procuring Contracting Officer, the Government technical activity designated in the contract, and the Government Quality Assurance Representative and include proposed corrective action whenever the following condition occurs: the contractor's Group A audit fails to clear for sampling, i.e. remains on 100% inspection, for 25 consecutive units. Notification is applicable only to units with major defects. For this audit the requirements of MIL-STD-105 shall apply, except for AQL's, and all electrical defects are considered major.

TABLE I

GROUP A INSPECTION SCHEDULE

INSPECTION	REQ'T PARA	INSP PARA
Mech & Visual	3.24	4.10
UHF AM CHANNELS		
Audio Output (RCVR)	3.18.1.1	4.7.5.1a
Sensitivity (RCVR)	3.18.1.2	4.7.5.1b
Distortion (RCVR)	3.18.1.3	4.7.5.1c
Power Output (XMTR)	3.18.1.5	4.7.5.1e
Percent Modulation	3.18.1.6	4.7.5.1f
Sidetone (XMTR)	3.18.1.7	4.7.5.1g
Distortion (XMTR)	3.18.1.8	4.7.5.1h
VHF AM CHANNELS		
Audio Output (RCVR)	3.18.2.1	4.7.5.2a
Sensitivity (RCVR)	3.18.2.2	4.7.5.2b
Distortion (RCVR)	3.18.2.3	4.7.5.2c
Power Output (XMTR)	3.18.2.4	4.7.5.2e
Percent Modulation (XMTR)	3.18.2.6	4.7.5.2f
Sidetone (XMTR)	3.18.2.7	4.7.5.2g
Distortion (XMTR)	3.18.2.8	4.7.5.2h
VHF-FM CHANNELS		
Audio Output (RCVR)	3.18.3.1	4.7.5.3a
Sensitivity (RCVR)	3.18.3.2	4.7.5.3b
Distortion (RCVR)	3.18.3.3	4.7.5.3c
Power Output (XMTR)	3.18.3.4	4.7.5.3e
Deviation (XMTR)	3.18.3.6	4.7.5.3f
Sidetone (XMTR)	3.18.3.7	4.7.5.3g
Distortion (XMTR)	3.18.3.8	4.7.5.3h

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TABLE I (Cont'd)

INSPECTION	REQ'T PARA	INSP PARA
HF-SSB CHANNELS		
Audio Output (RCVR)	3.18.4.1	4.7.5.4a
Sensitivity (RCVR)	3.18.4.2	4.7.5.4b
Power Output (XMTR)	3.18.4.3	4.7.5.4c
Percent Modulation (XMTR)	3.18.4.4	4.7.5.4d
Sidetone (XMTR)	3.18.4.5	4.7.5.4e
Radio Channel Crosstalk	3.18.5	4.7.5.5b
Volume Control Attenuation	3.18.6	4.7.5.5a
Foot Switch	3.18.7	4.7.5.5c
Voice Security	3.18.8	4.7.5.5d
Telephone System	3.18.9	4.7.7
Teletype System	3.18.10	4.7.6

4.5.2 Group B inspection. This inspection, including sampling, shall conform to TABLE II and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall be performed on centrals that have passed Group A inspection. For this inspection, one (1) central from every seven (7) on order shall be inspected. If a deficiency is found during Group B testing, testing for that deficient parameter shall be done on two (2) Centrals, one assembled immediately prior to and the other subsequent to the deficient Central. If no deficiencies are noted in the two additional Centrals, repair of the deficient Central should be made and a record of corrective action taken should be maintained and kept on file. If the deficiencies are also noted in the additional units tested, notification should be made to the procurement activity of actions necessary to correct the problem.

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TABLE II - GROUP B INSPECTION

INSPECTION	REQ'T PARA	INSP PARA
UHF AM CHANNELS		
VSWR (XMTR)	3.18.1.4	4.7.5.1d
VHF AM CHANNELS		
VSWR (XMTR)	3.18.2.5	4.7.5.2d
VHF FM CHANNELS		
VSWR (XMTR)	3.18.3.5	4.7.5.3d
SYSTEM TESTS		
Power Requirements	3.14	4.7.1
Controller Facilities	3.16	4.7.3
Signal Control Box Tests	3.17	4.7.4
Interchangeability	3.22	4.9

4.5.3 Group C inspection. This inspection shall be as listed in TABLE III and shall be performed on sample unit(s) that have been subjected to and met GROUP A and GROUP B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.3.1.

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TABLE III

GROUP C INSPECTION

	REQ'T PARA	INSP PARA
SUB-GROUP I		
Temperature High	3.19b	4.6b
Temperature Low	3.19c	4.6c
Altitude	3.19a	4.6a
Rain	3.19e	4.6e
Wind	3.19f	4.6f
Vibration	3.19j	4.6j
Bench Handling	3.19k	4.6m
SUB-GROUP II		
Humidity	3.19d	4.6d
Sand and Dust	3.19g	4.6g
Fungus	3.19h	4.6h
Water Fording	3.19i	4.6i
Shock Drop	3.19j	4.6k
Shock, Rail Impact	3.19j	4.6l
Storage	3.15	4.7.2
Electromagnetic compatibility	3.20	4.8

4.5.3.1 Sampling for inspection. Equipments shall be selected at random for each group of inspections in TABLE III as specified in 4.5.3.1.1 and 4.5.3.1.2. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. Selection of the samples shall be as specified and with 45 calendar days after selection, the samples shall have been subjected to and completed Group C inspections. (See 6.3)

4.5.3.1.1 Sub-Group I. One equipment (TSC-61) shall be selected from the first 20 shelters produced. For subsequent Group C inspections, one unit for each additional 75 units or fractions thereof shall be selected for inspection.

4.5.3.1.2 Sub-Group II. One equipment (TSC-61) shall be selected from the first 20 shelters produced. For subsequent Group C inspections one unit for each additional 125 units or fractions thereof shall be selected for inspection. Electromagnetic compatibility shall be performed as cited in 4.8.4.

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4.5.3.2 Non-compliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of Conforming Group C Sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

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TABLE IV

ORDER OF ENVIRONMENTAL AND STRESS INSPECTION

INSPECTION (NOTE 1)	NUMBER OF SAMPLES				
	ONE	TWO		THREE	
	Unit 1	Unit 2	Unit 1	Unit 2	Unit 3
NON-DAMAGING (NOTE 2)					
Temperature					
Low----------1	1		1		
High----------2	2		2		
Rain----------3	3		3		
Altitude (elevation)----------4		1		1	
Dust----------5		2			1
POTENTIALLY DAMAGING					
Humidity----------6 (moisture resistance)	4		4		
Fungus----------7 (NOTE 3)		7			2
Shock & Vibration----------9 (Munsen Road)		3			3
Shock drop----------10		4	5		
Shock, rail impact----------11		5	6		
Water Fording----------12		6	7		
*Vibration, including----------1 resonance search-or fatigue type vibration		1	1		
*Shock, bench handling----------2		2	2		

NOTE 1: Other inspections may precede, follow, or be interspersed between the non-damaging tests.

NOTE 2: The order on non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient, except that the vibration test shall be performed on the same unit used for the bounce and shock tests, and, if an immersion test is required after vibration or shock tests, the test shall be performed on the unit used for the initial immersion test.

NOTE 3: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before preceeding with subsequent tests.

* Applicable only to items manufactured by the contractor.

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4.6 Test procedure.

a. Altitude. (3.19a) The assembly shall be subjected to the test of Method 500, Procedure I, Standard MIL-STD-810.

b. High temperature. (3.19b) The assembly shall be subjected to the test of Method 501, Procedure II, of MIL-STD-810. For Step 4 the chamber temperature shall be 71°C (160°F). Steps 7 and 8 shall be performed as follows:

(1) Raise chamber temperature to 71°C (160°F) and maintain until temperature stabilization of the assembly. Initiate operation of those externally mounted electronic equipments of the assembly which do not employ natural or forced air cooling in their design. These items shall comply with full operational performance requirements within a time period of 30 minutes. Operation shall continue until test item stabilization with no degradation in performance.

(2) Within a time period of 10 minutes, lower the chamber temperature to 52°C (125°F) and, with the exception of the air conditioning system (if any), initiate operation of the assembly as desired so as to attain full electronic operational performance within a time period of 30 minutes. Assembly operation shall be initiated immediately upon the chamber temperature reaching 52°C, and the 30 minute period shall be measured from the time assembly operation is commenced. Operation shall continue until assembly stabilization with no degradation in performance.

c. Low temperature. (3.19c) The assembly shall be subjected to the test of Method 502, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -57°C (-70°F) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (Step 4) shall be -46°C (-50°F). In Step 5, the assembly shall be turned "on" including electronic equipment and assembly accessories, heating system, etc., as required for normal operation and shall comply with full operational performance requirements within a time period of 30 minutes.

d. Humidity. (3.19d) The assembly shall be subjected to the test of Method 507, Procedure II of MIL-STD-810. During Steps 3 and 7, the shelter or van doors, windows, louvers and vents shall be open. Prior to measurements, excess surface water may be removed from the equipment by wiping external surfaces only. In Step 6, measurements shall be taken during the last 5 hours of the last cycle.

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e. Rain. (3.19e) The assembly shall be subjected to the test of Method 506, Procedure I, of MIL-STD-810. During each 30 minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period as specified in the procedure. The equipment shall be operated during the last 10 minutes of the 30 minute period.

f. Wind load. (3.19f) Since there are numerous ways to determine compliance with wind loading requirements, and methods used must be tailored to specific test items, it is expected that the equipment development contractor will devise and submit a proposed procedure to the procurement agency for approval. Acceptable methods include the employment of wind tunnel facilities, submission of calculations utilizing established data, actual loading of the test item with weights and pulleys or other suitable arrangement, and any combination of the above. (NOTE: If it can be determined that the nature of the equipment and/or it's application is such that wind load effects would be negligible, testing may be omitted.)

g. Sand and dust. (3.19g)

(1) Individual external equipments of the assembly shall be tested as required to demonstrate the resistance of the assembly to blowing sand and dust. Such components shall be subjected to the test of Method 510, Procedure I, of MIL-STD-810.

(2) The contractor-furnished internal equipment shall be subjected to the test of Method 510, Procedure I of MIL-STD-810, with the exception that the air velocity in Steps 1, 2 and 3 shall be 200 $\pm$ 100 fpm. Step 3 may proceed immediately after stabilization in Step 2 if desired.

h. Fungus. (3.19h) Individual electronic equipments of the assembly shall be tested as required to demonstrate resistance to fungus growth. Such items shall be subjected to the test of Method 508, Procedure I, of MIL-STD-810. Gasket sealed or other non-hermetically sealed test items shall be opened during test exposure and all internal surfaces shall be sprayed with spore suspension as well as external surfaces. Operation of the equipment is not required but examination at the conclusion of the exposure period shall reveal no evidence of viable fungus on any surfaces.

i. Water fording. (3.19i) The assembly shall be subjected to the water fording test contained in the specification for the basic shelter used in the assembly construction. This test may be omitted if the shelter has not been modified below the applicable water line.

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j. Vibration. (3.19j) The assembly shall be tested as follows:

(1) Munson road test. Transported by its intended vehicle, the assembly shall be driven five times over the following sections of the Munson Test Course at Aberdeen Proving Ground, Maryland, at the individual speeds and in the order listed:

Course washboard (6 inch waves spaced 75 inches apart)	5 mph
Belgian block	20 mph
Radial washboard (2 inch waves to 4 inch waves)	15 mph
Single corrugations (4 inch to 6 inch waves)	20 mph
Any short sections between the above	20 mph

At the conclusion of the test the assembly shall best the specified performance requirements and have no physical damage.

(2) Resonance search. The contractor-furnished internal equipment shall be subjected to the test of Method 514, Procedure IX of MIL-STD-810.

k. Shock (drop). (3.19j) The assembly shall be subjected to the test of Method 516, Procedure II, of MIL-STD-810. The test shall be applied to a shelter assembly in the same manner as it is applied to a transit cased package of the applicable weight class.

l. Shock (rail impact). (3.19j) The assembly shall be subjected to the rail humping (impact) test contained in the specification for the basic shelter used in the assembly construction.

m. Shock (bench handling). (3.19k) The contractor-furnished internal equipment shall be subjected to the test of Method 516, Procedure V of MIL-STD-810.

4.7 System inspections.

4.7.1 Power requirements test. Tests shall be made to verify that the requirements of 3.14.1 through 3.14.7 have been met.

4.7.2 Storage of operation components during transit. Verify that the requirements of 3.15 have been met.

4.7.3 Controller communications facilities. Tests shall be made, which are acceptable to the Government, to verify that the requirements of 3.16.1 through 3.16.5 have been met.

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4.7.4 Signal control box. A test shall be made, which is acceptable to the Government, to verify that the interface circuits between the radios and controller consoles perform as specified in 3.17.1 through 3.17.3.

4.7.5 Radio system tests. Three channels for each radio, randomly selected, shall be tested to show compliance with 3.18 through 3.18.9.4.

4.7.5.1 UHF-AM channels. For each of the UHF-AM channels selected in 4.7.5 conduct the following tests:

a. Receiver audio output. Adjust all receivers for unsquelched condition. Set RF input to 1,000 microvolts modulated 30% at 1,000 Hz. Adjust all volume controls for maximum output. Check for a minimum of 230 mW output from each audio control box in accordance with 3.18.1.1.

b. Receiver sensitivity. Set RF input to 4 microvolts with signal modulated 30% at 1,000 Hz. Adjust volume control on "Radio Key Panel" to 100 milliwatts. Observe output with and without modulation. Record the difference in "dB" to show compliance with 3.18.1.2.

c. Receiver distortion. Set RF input to 1,000 microvolts, modulated 30% at 1,000 Hz. Adjust volume control output for 100 milliwatts. Check distortion at each operators position to determine compliance with 3.18.1.3.

d. Transmitter VSWR. Connect thru-line wattmeter and antenna to AN/ARC-51BX. Record forward and reflected RF power output. Calculate VSWR to show compliance with 3.18.1.4.

e. Transmitter power output. Connect RF power output meter with dummy load to AN/ARC-51BX output antenna connection, key transmitter. Measure output power to show compliance with 3.18.1.5.

f. Transmitter modulation level. Apply standard 1,000 Hz test signal of 0.6 millivolt to the microphone input of the audio control box, key transmitter and measure percent modulation at the antenna entry panel to show compliance with 3.18.1.6.

g. Transmitter sidetone. Applying the standard input signal (1,000 Hz 0.6 millivolt) measure sidetone level at the headset output of the audio control box to show compliance with 3.18.1.7.

h. Transmitter distortion. Using a linear detector, and the standard test signal (1,000 Hz 0.6 millivolt) applied to the microphone input, measure the detected audio output distortion at the antenna entry panel in accordance with 3.18.1.8.

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4.7.5.2 VHF-AM channels. For each of the VHF-AM channels selected in 4.7.5, conduct the following tests:

a. Receiver audio output. Adjust all receivers for unsquelched condition. Set RF input to 1,000 microvolts modulated 30% at 1,000 Hz. Adjust all volume controls for maximum output. Check for a minimum of 230 mW output from each audio control box in accordance with 3.18.2.1.

b. Receiver sensitivity. Set RF input to 3.0 microvolts with signal modulated 30% at 1,000 Hz. Adjust volume control on "Radio Key Panel" to 100 milliwatts. Observe output with and without modulation. Record the difference in "dB" to show compliance with 3.18.2.2.

c. Receiver distortion. Set RF input to 1,000 microvolts, modulated 30% at 1,000 Hz. Adjust volume control output for 100 milliwatts. Check distortion at each operators position to determine compliance with 3.18.2.3.

d. Transmitter VSWR. Connect thru-line wattmeter and antenna to AN/ARC-73. Record forward and reflected RF power output. Calculate VSWR to show compliance with 3.18.2.5.

e. Transmitter power output. Connect RF power meter with dummy load to AN/ARC-73 output "Ant." connection. Key transmitter. Measure output power to show compliance with 3.18.2.4.

f. Transmitter modulation level. Apply standard 1,000 Hz test signal of 0.6 millivolt to the microphone input of the audio control box, key transmitter and measure percent modulation at the antenna entry panel to show compliance with 3.18.2.6.

g. Transmitter sidetone. Applying the standard input signal (1,000 Hz 0.6 millivolt) measure sidetone level at the headset output of the audio control box to show compliance with 3.18.2.7.

h. Transmitter distortion. Using a linear detector and standard test signal (1,000 Hz 0.6 millivolt) applied to the microphone input measure the detected audio output distortion at the antenna entry panel in accordance with 3.18.2.8.

4.7.5.3 VHF-FM channels. For each of the VHF-FM channels selected in 4.7.5, conduct the following tests:

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a. Receiver audio output. Adjust all receivers for unsquelched condition. Set RF input to 1,000 microvolts ± 8 kHz deviation at 1,000 Hz. Adjust all volume controls for maximum output. Check for a minimum of 230 mV outputs from each audio control box in accordance with 3.18.3.1.

b. Receiver sensitivity. Set RF input to 1.5 microvolts using ± 8 kHz deviation at 1,000 Hz. Adjust volume control on "Radio Key Panel" to 100 milliwatts. Observe output with and without modulation. Record the difference in "dB" to show compliance with 3.18.3.2.

c. Receiver distortion. Set RF input to 1,000 microvolts, deviation ± 8 kHz at 1,000 Hz. Adjust control panel output to 100 milliwatts. Check distortion at each operators position, to determine compliance with 3.18.3.3.

d. Transmitter VSWR. Connect thru-line wattmeter and shelter antenna to VRC-46. Record forward and reflected RF power output. Calculate VSWR to show compliance with 3.18.3.5.

e. Transmitter power output. Connect RF power meter with dummy load to AN/VRC-46 output "Ant." connection. Key transmitter. Measure output power to show compliance with 3.18.3.4.

f. Transmitter deviation level. Apply standard 1,000 Hz test signal of 0.6 millivolt to the microphone input of the audio control box, key transmitter and measure deviation of transmitter at the antenna entry panel to show compliance with 3.18.3.6.

g. Transmitter sidetone. Applying the standard input signal (1,000 Hz 0.6 millivolt) measure sidetone level at the headset output of the audio control box to show compliance with 3.18.3.7.

h. Transmitter distortion. Using the detected audio output of a deviation meter and the standard test signal (1,000 Hz 0.6 millivolt) applied to the microphone input measure the detected audio output distortion at the antenna entry panel in accordance with 3.18.3.8.

4.7.5.4 HF-SSB channels. For each of the HF-SSB channels selected in 4.7.5 conduct the following tests:

a. Audio output.

(1) Remove antenna lead from coupler. Connect "Signal Generator" RF output to "Aux. Ant." input of AN/ARC-102. Set function switch on control head to "AM" and adjust frequency to 7.3 MHz. Adjust "Sensitivity" control to maximum and adjust volume on all "Radio Key Panels" to maximum. Set "Audio Wattmeter" to 8 ohms.

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(2) Set "Signal Generator" to 1,000 microvolts, modulated 30% at 1,000 Hz. Adjust frequency to 7.3 MHz until maximum audio is obtained. Check for a minimum of 230 mV output from each audio control box in accordance with 3.18.4.1.

b. Sensitivity. Repeat above test. Set RF output of "Signal Generator" to 3.0 microvolts, modulated 30% at 1,000 Hz. Adjust volume control on "Radio Key Panel" to 100 milliwatts (20 dB) on the output of "Audio Wattmeter". Observe output with and without modulation. Record the difference in "dB", to show compliance with 3.18.4.2.

c. Transmitter power output (AM). Turn off audio oscillator and key transmitter. Measure output power to show compliance with 3.18.4.3.

d. Transmitter modulation level. Apply standard 1,000 Hz test signal of 0.6 millivolt to the microphone input of the audio control box, key transmitter and measure percent modulation at the antenna entry panel to show compliance with 3.18.4.4.

e. Transmitter sidetone. Applying the standard input signal (1,000 Hz 0.6 millivolt) measure sidetone level at the headset output of the audio control box to show compliance with 3.18.4.5.

4.7.5.5 Voice communication system. Conduct the following test on the voice communication system:

a. Attenuation (volume control). Adjust modulated RF input to 1,000 microvolts into any receiver and, with audio control box volume adjusted to maximum, set radio set volume control for 200 milliwatts at output of control panel (8 ohms). Rotate volume control (operator's position) to minimum. Measure output change (in dB) at each of the operator's positions in accordance with 3.18.6

b. Radio channel cross talk. Maintain the 200 milliwatt obtained in above test. Switch off that receiver and, one at a time, switch in each of the nine remaining receivers (all squelched) and record the difference in level related to 200 milliwatt. Repeat this test for each of the operator's control positions. The requirements of 3.18.5 shall be met.

c. Foot switch. Check operation of foot switch for compliance with 3.18.7.

d. Voice security. The security circuit shall be checked to ascertain that it meets requirements delineated in 3.18.8 thru 3.18.8.6.

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4.7.6 Teletype system. An operational test shall be made to determine if performance of the teletype system is in accordance with 3.18.10 thru 3.18.10.2.4.

4.7.6.1 Radio teletype transmitter power output. The radio teletype power output shall be tested in accordance with 3.18.10.2.2.

4.7.6.2 Radio teletype receiver sensitivity. The radio receiver shall be tested to determine compliance with 3.18.10.2.3.

4.7.6.3 The teletype system shall be checked for 60 words per minute operated as required by 3.18.10.2.4.

4.7.7 Telephone system test. Conduct the following tests on the telephone system:

a. Telephone audio input. With 1,000 Hz signal at zero dBm at the telephone entrance panel, the audio level at the headset output of the audio control box shall be measured to demonstrate compliance with 3.18.9.1.

b. Telephone audio output. With a 1,000 Hz signal at a level of 0.6 millivolts at the microphone input, the output level at the entry panel binding posts shall be -6 dBm to +3 dBm, demonstrating compliance with 3.18.9.2.

c. Crosstalk test (telephones). With all the telephone controls set for normal operation, measure the crosstalk between telephone lines to demonstrate compliance with 3.18.9.3.

d. Night alarm buzzer. Apply a ring-in signal to any one of the telephone lines and then stop the ring-in signal. Observe that the buzzer shall buzz until the lever switch for that line is moved to the talk position. The requirements of 3.18.9.4 shall be met.

4.8 Electromagnetic compatibility. The AN/TSC-61( ) as fabricated in accordance with this specification, shall be tested for compliance as follows herein.

4.8.1 System Compatibility. The total system shall be tested for compliance to the requirements of 3.20.1 of this specification. Test set-ups and measurement techniques shall be in accordance with MIL-STD-462.

4.8.2 Contractor-furnished equipment. Contractor-furnished equipments and vendor supplied items shall be tested for compliance to the requirements of 3.20.2 of this specification. Measurement techniques shall be in accordance with MIL-STD-462.

4.8.3 Intra-system compatibility. The AN/TSC-61( ) shall be tested for compliance to the requirements of 3.20.3. Tests shall be performed, with all antennas installed, as follows:

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a. Transmitters vs receivers (harmonics, spurious, image frequency, intermodulation).

b. Receiver vs receiver interaction.

c. Receiver desensitization.

d. Transmitters vs flight status consoles.

e. Air conditioner vs receiver.

f. Air conditioner vs flight status console.

g. Air conditioner vs teletype printer.

h. Heater vs receiver.

i. Heater vs flight status consoles.

j. Heater vs teletype printer.

4.9 Interchangeability. The following shall be gaged, manual and/or physically demonstrated to determine conformance to the physical interchangeability requirements of 3.22. When a listed item is not within specified or design limits it shall be a major defect:

(a) External and internal dimensions of cases, covers and insertable assemblies.

(b) Dimensions of mounting racks where units are located,

(c) Location of hinges and fasteners which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fastener-slides, racks and mountings which receive mating parts units.

4.10 Visual and Mechanical inspection. Centrals shall be examined for defects listed in MIL-STD-252. The requirements of 3.24 shall be met.

4.11 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with Section 5.

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5. Preparation for delivery.

5.1 Preservation, packaging, packing and marking shall be in accordance with Specification MIL-S-55505.

6. NOTES

6.1 Intended use. Flight Coordination Central AN/TSC-61( ) provides facilities for air traffic coordination, warning air defense identification and in-flight assistance with an assigned zone of responsibility on a continuous basis. Altitude, time and distance flight plan data for airborne aircraft can also be coordinated. It provides facilities to establish air-to-ground radio communications with frequency modulated equipment in the very high frequency range and with amplitude-modulated (AM) equipment (pilot command control) in the ultra-high frequency, very high frequency and high frequency (HF) ranges. Facilities are also included for telephone, radio-teletype for ground-to-ground radio communications with associated air fields and ground installations.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required.

c. Level of packaging and level of packing required for shipment. (Level A, Level B, or Level C.)

d. Preproduction inspection:

(1) Two (2) sample units of each equipment (TSC-61) generally required so that lengthy environmental tests can be completed on specified sample units while complete performance measurements can be made on one sample unit. (See 3.3)

(2) First Article pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon. (Including rough handling).

e. Marking and shipping of sample.

f. Place of final inspection.

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g. Technical literature required.

h. Quantity of spare parts required.

6.3 Group C inspection. Approval to ship unit(s) on order may be withheld, at the discretion of the Government pending decision from the procuring activity, until compliance with 4.5.3.1 is complete.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/TSC-61W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.5 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in the specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian
Army EL

Preparing Activity
Army EL
Project No. 5895-A166

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-F-55356(EL) FLIGHT COORDINATION CENTRAL AN/TSC-61( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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